KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRONICS

DATE: 26th July 2019	TIME: 3 HOURS (9:00AM – 12:00 NOON)

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION ONE: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The $\frac{dv}{dt}$ effect in a SCR can result in
   a) High rate of rise of anode voltage
   b) Increased junction capacitance
   c) False triggering
   d) Low capacitive charging current
2. Failures that could be anticipated by prior examination is called
   a) Gradual failure
   b) Sudden failure
   c) Degradation failure
   d) Catastrophic failure
3. The 8086 microprocessor fetch instruction one after another from _____ of memory
   a) Data segment
   b) Code segment
   c) Extra segment
   d) Stack segment
4. When a thyristor is negatively biased
   a) All the three junctions are negatively biased
   b) Outer junctions are positively biased and the inner junction is negatively biased
   c) Outer junctions are positively biased and the inner junction is positively biased
   d) The junction near the anode is negatively biased and the one near the cathode is positively biased
5. If A and B are the inputs of a half adder, the sum is given by
   a) A AND B
   b) A XOR B
   c) A OR B
   d) A EX – NOR B
6. A semiconductor memory device in which the stored data will remain permanently stored as long as power supply is present without the need of periodically rewriting the data into memory is called
   a) Static memory
   b) Sequential access memory
   c) Dynamic memory
   d) Read only memory
7. The register which is a source of data for the arithmetic logic unit and all other registers are called
   a) General purpose register
   b) Status register
   c) Accumulator
   d) Flag registers
8. Ramp ADC (Analog to Digital Converter) to generate a
   a) Ramp test waveform
   b) Ramp equation
   c) Ramp input

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d) Ramp output

9. A zener diode utilizes _____ characteristics for voltage regulation
   a) Forward
   b) Reverse
   c) Both forward and reverse
   d) None of the above
10. Which of the following is Non- volatile storage
   a) Backup
   b) Primary
   c) Secondary
   d) Cache
11. The main reason why a bleeder resistor is used in a dc power supply is that it
   a) Keeps the supply on
   b) Improves voltage regulation
   c) Improves filtering action
   d) Both b and c
12. In a system the average time to repair any fault is two hours. What is maintainability for a time of four hours
   a) 63.8%
   b) 8.65%
   c) 86.5%
   d) 6.38%
13. The external chips of a microprocessor initiate the following operation except;
   a) Reset
   b) Storage
   c) Interrupt
   d) Hold
14. The most popular systematic fault location method is;
   a) Input to output
   b) Output to input
   c) Half split
   d) All of the above
15. In a rectifier, larger the value of a shunt capacitor filter
   a) Larger the p-p value of ripple voltage
   b) Larger the peak current in the rectifying diode
   c) Longer the time that current pulse flows through the diode
   d) Smaller the DC voltage
16. The purpose of a comparator is to
   a) Amplify an input voltage
   b) Detect the occurrence of a changing input voltage
   c) Maintain a constant output when the dc input voltage changes
   d) Produce a change in input voltage when an input voltage equals the reference voltage
17. A 4 bit R/2R digital to analogue (DAC) converter has a reference of 5 volts. What is the analogue output for the input code 0101
   a) 0.3125v
   b) 3.125v

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c) 0.78125v
d) >3.125v

18. The purpose of a clock input to a flip flop is to
a) Clear the device
b) Set the device
c) Always cause the output to change the states
d) Cause the output to assume a state dependent on the controlling unit

19. According to the distributive laws of Boolean algebra, expression $(A + \overline{A}.B)$ equals to
a) A + B
b) B
c) AB
d) A

20. In an inverting amplifier, the two input terminals of an ideal OP-AMP are at the same potential because
a) The two input terminals are directly shorted internally
b) The input impedance of the OP-AMP is infinity
c) Common – mode rejection ratio is infinity
d) The open-loop gain of the OP-AMP is infinity

21. The number 17_8 is equivalent to binary
a) 111_2
b) 1110_2
c) 10000_2
d) 1111_2

22. A Hartley oscillator is used for generating
a) Very low frequency oscillation
b) Radio frequency oscillation
c) Microwave oscillation
d) Audio frequency oscillation

23. Optical couplers are designed to _____ one circuit from another
a) Control
b) Isolate
c) Disconnect
d) Protect

24. In Excess – 3 code each coded number is _____ than in BCD code
a) Four larger
b) Twice larger
c) Three larger
d) None of the above

25. A multimode step index fiber has a large core diameter of range
a) 100 to 300 μm
b) 100 to 300 nm
c) 200 to 500 μm
d) 200 to 500 nm

26. RC coupling is popular in low – level audio amplifier because it
a) Has better low frequency response
b) Is inexpensive and needs no adjustments

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c) Provides an output signal in phase with the input signals
d) Needs low voltage battery for collector supply

27. Class _____ amplifiers are normally operated in a push pull configuration in order to produce an output that is replica of the input
a) A
b) B
c) C
d) AB

28. Which among the following is provided by an optical receiver for the regeneration of data signal with minimum error
a) Photo-diode
b) Signal processing circuit
c) Linear circuitry
d) None of the above

29. Laser beam is made of
a) Electrons
b) Highly coherent photon
c) Elastic particles
d) Excited atoms

30. Which of the power amplifiers is not intended primarily for large signal or power amplification
a) Class A
b) Class B or AB
c) Class C
d) Class D

31. The design of Arithmetic logic unit is based on
a) Sequential logic
b) Combinational logic
c) Multiplexing
d) Binary circuits

32. The basic building blocks of the arithmetic unit in a digital computer are
a) Subtractors
b) Adders
c) Multiplexer
d) None of the above

33. A CRO can be used to measure
a) A.C voltages only
b) D.C voltages only
c) Frequency
d) Any of the above

34. The multi vibrator which generates square wave of its own is the _____ multi vibrator
a) Mono stable
b) Bi stable
c) A stable
d) None of the above

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35. To get an output from an XNOR gate, its both inputs must be
    a) High
    b) Low
    c) At the same logic level
    d) At the opposite logic levels
36. A logic circuit that accepts several data inputs and allows only one of them at a time to get through at the output is called
    a) Decoder
    b) Encoder
    c) De multiplexers
    d) Multiplexers
37. When the temperature increases, the inter base resistance (R_{BB}) of a UJT
    a) Increases
    b) Decreases
    c) Remain the same
    d) Fluctuates
38. When the input of a NAND gate are connected together. The resulting circuit is
    a) OR gate
    b) AND gate
    c) NOT gate
    d) None of the above
39. Using 2's complement, subtract 1101_2 from 1010_2
    a) 0011_2
    b) -0011_2
    c) 1100_2
    d) -1100_2
40. In LED light is emitted because
    a) Recombination of charge carriers takes place
    b) Diode gets heated up
    c) Light falling on the diode gets amplified
    d) Light get reflected due to lens action

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SECTION TWO: SHORT ANSWER QUESTIONS (40 MARKS)

1. State five ways used for firing a SCR (5 marks)
2. Differentiate between combinational circuit and sequential circuit (5 marks)
3. Define the following terms as they are applied in semiconductor memories (5 marks)
   i. Byte
   ii. Memory cell
   iii. Address
   iv. Memory word
   v. Word size
4. Evaluate the following operations
   i. $10011_2 - 10100_2$ using 2's complement (3 marks)
   ii. 175_{10} into its octal equivalent (2 marks)
5. With aid of a diagram explain the operation of a photo-diode (5 marks)
6. a) Define the term flip flop (1 marks)
   b) Explain the performance criteria for both ADC (Analogue to digital converter) and DAC (Digital to analogue converter) (4 marks)
7. A power supply, oscillator and amplifier are all used in a simple system. Calculate the individual and system reliabilities for 1000 hours operating period, if the MTBFs are 20,000 hours, 100,000 hours and 50,000 hours respectively (5 marks)
8. Explain the function of the following registers in a microprocessor
   i. Instruction register (2.5 marks)
   ii. Status register (2.5 marks)

SECTION THREE: LONG ANSWER QUESTION (20 MARKS)

1. a) Explain the order of operation by microprocessor using the fetch execute cycle flow chart (6 marks)
   b) An amplifier has an open loop gain Av = 100,000. A negative feedback of 10db is applied, find
      i. Voltage gain with feedback (2 marks)
      ii. Value of feedback fraction Mv (3 marks)
   c) With the aid of a block diagram of a signal conditioning system explain the operation of each block (7 marks)
   d) Explain the meaning of toggling as used in digital electronics (2 marks)

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